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Strategic environmental assessment—exploring connections between environmental and health issues

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Abstract

In this paper, how changes to environmental aspects considered in strategic environmental assessment (SEA) can affect human health is explored, based on a comprehensive and systematic literature review. Whilst health is a key aspect for consideration in SEA, how it should be approached has remained subject to discussion. There is agreement, though, that at a minimum, health impacts from bio-physical environmental changes need be assessed. What this means is elaborated on in this paper. Challenges to and opportunities for an effective consideration of health in SEA are briefly outlined.

Introduction

Strategic Environmental Assessment (SEA) aims at effectively integrating environmental considerations into policy, plan, programme (PPP), and other strategic decision-making processes. SEA complements Environmental Impact Assessment (EIA) applied to individual projects and aims at establishing the cumulative, synergistic, and long-term impacts of PPPs and other strategies on environmental issues that are either not or only insufficiently considered in project-level assessments (Fischer 2007). By establishing potential negative impacts as well as opportunities for enhancing positive outcomes at a time when concrete development projects are yet to be decided on, SEA can function as a key mechanism for supporting sustainable development, helping to mitigate adverse environmental impacts, whilst pro-actively supporting positive outcomes (Fischer and Gonzalez 2021).

By 2025, SEA had been statutorily implemented in more than 100 countries (González and Fischer 2026). Probably the most widely known SEA regulation internationally is the European SEA Directive from 2001 (EC 2001). In terms of international SEA requirements, the United Nations Economic Commission for Europe SEA Protocol to the Espoo Convention on EIA in a Transboundary Context (UNECE 2003) is also a widely acknowledged and important document. It is of relevance for 56 countries in Europe, Central Asia and North America.

Whilst human health is a key aspect for consideration in SEA (Fischer et al. 2008), there is currently no consensus on how exactly it should be integrated. A starting point for debate can be the most commonly used definition for health from the 1946 constitution of the World Health Organization (WHO):

> "Health is a state of complete physical, mental and social well-being and not merely the absence of disease or infirmity" (World Health Organization 2020).

The importance for considering health in SEA is derived from an ability of underlying PPPs and other strategies to affect human health (direct impact) and to influence human behaviour (indirect impact), which can lead to improved health outcomes. Regarding direct impacts, issues such as noise and pollution are relevant. With regards to indirect

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impacts, as an example, built and natural environments can either encourage or discourage certain activities, such as cycling or walking to work.

It is important that SEA supports alignment with the United Nations 2030 Agenda (United Nations 2015). This consists of 17 Sustainable Development Goals (SDGs) goals and 169 targets. In the context of the SDGs, Menne et al. (2020) noted how health is both an enabler and a major outcome of sustainable development. With regards to health, SDG 3 on good health and well-being, SDG 10 on reduced inequalities, and SDG 11 on sustainable cities and communities are of particular importance. In this context, SEA enables the anticipatory and intersectoral consideration of social and environmental aspects that are interconnected with health, influencing decisions in sectors such as transport, housing, and energy.

A key question of the current debate amongst practitioners and policy makers is to what extent the full range of health determinants (those that are seen to be at the heart of health impact assessments—HIAs; see e.g. Winkler et al. 2021) should also be fully included in SEA. Following WHO (2024), the determinants of health are income and social status, education, the physical environment, social support networks, genetics, health services, and gender. Whilst a wide range of these determinants can potentially be influenced by SEA (Carmichael et al. 2019), some have suggested that the starting point for including health should be the bio-physical environment (UNECE 2022).

It is not the aim of this paper to provide an answer to the question of whether all health determinants should always be considered in SEA. However, even if the focus was on the bio-physical environment only, what aspects should be considered remains unclear, and this is the starting point for this paper. The interlinkages between aspects of the bio-physical environment and human health was a key objective of the 'Pro Health' project conducted for the Irish Environmental Protection Agency (EPA) in 2024 to 2025 (https://www.ucd.ie/healthsea/). What follows in this paper is based on the research done for that project.

Health in SEA: international requirements

The UNECE SEA Protocol (UNECE, 2003) underlines the need for an integration of environmental and health considerations into PPP making processes. Article 1 of the Protocol highlights potential effects of PPPs on human health, alongside other environmental receptors that have been considered traditionally in environmental assessments, including e.g. air, water, and biodiversity. Importantly, the Protocol also calls for an identification of health inequities in particular areas, as well as vulnerable populations on the other. In this context, the Protocol advises that SEA should contribute to equitable outcomes.

The WHO has been instrumental in advancing the consideration of human health within SEA. In this context, the 'one health' framework plays a key role for identifying potential health risks associated with ecosystem degradation and zoonotic diseases (WHO, 2023). Besides COVID-19, viruses that have been passed from animals to humans also include others, e.g. Lyme Disease and Ebola. This is of relevance for SEA in the context of e.g. developments that are planned close to natural areas, undisturbed by human activity. Here, SEA should critically assess the possibility of zoonotic disease spread. Also, SEA can advise on pandemic preparedness measures.

In a European context, awareness for the importance of the integration of health into SEAs has gradually increased. A few countries have introduced guidelines and/or recommendations, including the UK (England and Scotland), Georgia, France and Ireland (Roue de Gall et al. 2024; Pyper et al. 2021; Scottish Environment Protection Agency 2019; Roue de Gall et al. 2014; Williams and Fisher 2007).

Subsequently, in this paper, the connections between bio-physical environmental factors to be considered in SEA, according to European Directive 2001/42/EC, and health are explored. Whilst there are other pathways and mechanisms through which human health may be affected by the implementation of PPPs, as indicated above, bio-physical environmental-health interconnections should always be a key concern in SEA. Importantly, as outlined in the introduction, EC and UNECE guidance documents refer to human health without limiting this to environmental factors. Therefore, other health aspects may need to be considered in SEA. However, these are not subject to this paper.

The need for a proportionate approach

SEA needs to balance the comprehensive assessment of health impacts with the practicalities and limitations of PPP making, as well as with the need to assess a range of environmental aspects. This means that not all aspects will and should always be considered. The International Association for Impact Assessment (IAIA)/European Public Health Association (EUPHA) reference paper for health in Environmental Impact Assessment (EIA) (Cave et al. 2020) has proportionality as one of its principles. Its focus is on the scoping stage and on identifying likely and significant impacts.

To assess impacts proportionally, the context of SEA needs to be understood. Whilst the scope of assessment should be set in consultation with relevant authorities, the specific decision tier (i.e. whether it is e.g. a policy, a plan or a programme) and the administrative level (whether the underlying PPP is prepared at e.g. national, regional or local

levels) play key roles when deciding on an adequate scope, for example with regards to not repeating data collection processes, assessing the same alternatives or unnecessarily assessing aspects that are not relevant (Fischer 2006; González et al. 2015). With regard to the choice of proportionate impacts, aspects and alternatives, the sector of application is also of importance. Finally, when attempting to devise proportionate SEA, the decision framework within which a particular PPP is prepared and the institutional capacity for conducting SEA (and the distribution of power of different institutional players, which is likely to influence decisions) should be taken into account in order to be able to e.g. counter biases. Figure 1 provides an example for how proportionate SEA (here called 'SEA format') can be determined by various contextual factors.

Source:* Fischer 2014, *p32

The principle of proportionality has been included in approaches towards SEA— and also EIA—in a few countries. In France, for example, the 2019 guidance 'The principle of proportionality in environmental assessment' recommends that the scope of assessment be adjusted to the relevance of the plan and its expected effects, including on human health. It outlines criteria such as the environmental sensitivity of the area, the nature of planned interventions, and the anticipated impacts, reinforcing the importance of tailoring the assessment to the specific context (Commissariat général au développement durable 2019).

It is important that even if only physical environmental determinants of health were to be considered in SEA, there would still be numerous potential interrelationships that need to be explored. This is why in this paper, health impacts associated with bio-physical environmental issues are elaborated on. This is intended to help those engaging with health in SEA to adequately address the issue.

Growing concerns on environmental-health interactions

There are currently growing concerns with regards to human health impacts from changes to the bio-physical environment, as well as the health consequences of human impacts on the natural environment. This is associated with various challenges, e.g. the climate change crisis and the recent COVID-19 pandemic. Associated discourses include those surrounding the 'environmental health' and 'one health' concepts. 'Environmental health' focuses on the impact of people on the environment and the impact of the environment on them (Moeller 2005). In 1995, the European Environment Agency defined environmental health as including:

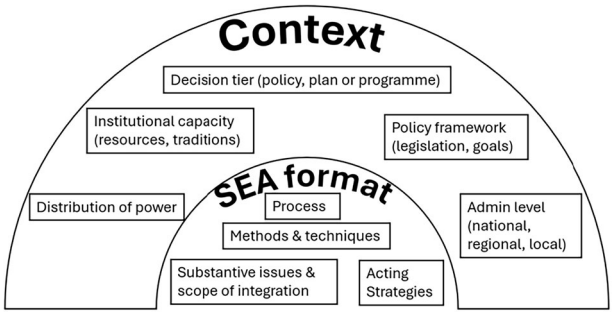

Fig. 1 The format of – proportionate – SEA as determined by contextual factors

> "… both, the direct pathological effects of chemicals, radiation and some biological agents, and the effects (often indirect) on health and well-being (see below) of the broad physical, psychological, social and aesthetic environment, which includes housing, urban development, land use and transport" (European Environment Agency 1995).

Major environmental health risk concerns include exposure to environmental pollutants (e.g. air and noise pollution, heavy metals) and climate-related events (European Environment Agency, 2023). Importantly, the environmental health discourse emphasises the preservation of the natural, living, and occupational (physical and social) environment. Therefore, it is highly relevant for SEA.

Secondly, 'one health' is:

> "an integrated, unifying approach to balance and optimise the health of people, animals and ecosystems" (World Health Organization 2023).

It follows the thinking around environmental health and emphasises the interconnections of human and animal health, acknowledging that human activities and stressed ecosystems provide opportunities for diseases to transmit to humans and spread (World Health Organization 2023). The thinking surrounding 'one health' is said to be key for preventing the next pandemic (European Environment Agency 2020; World Bank 2022).

Zoonotic diseases that are transmitted from animals to humans are viewed as a major threat, and more than 60% of emerging infectious diseases (EIDs) that are reported globally come from animals (World Health Organization 2023). As such, human activities that interfere with wildlife are of fundamental concern. These activities include e.g. agriculture, forestry and urban sprawl (World Bank 2022). SEA plays a key role in assessing impacts in associated PPPs. Therefore, it is important to gather evidence for bio-physical environmental–health interactions, as is done by this paper.

Literature review methodology

There is a long history of doing research on the relationship between the environment and human health. In this context, a simple desktop search on the Scopus database (January 2024) using the keywords 'human' AND 'health' AND 'environment' returned a total of 78,741 documents. These documents are from various academic disciplines.

The decision to use Scopus for identifying relevant publications for exploring environmental and health interactions was taken due to this being a comprehensive database of peer-reviewed journals in the field of physical and social sciences, with publications from the past several decades being listed. Based on other research done in the field of environmental assessment, Scopus was considered the most adequate of the available options (see e.g. Fischer 2023). There are other databases that were discarded. The Web of Science, for example, is more restrictive than Scopus, in particular with regard to the social sciences and humanities. However, many relevant publications on SEA are from these disciplines. Also, Google Scholar was considered. This probably includes the widest range of academic and non-academic documents. However, anyone can put any document into Google Scholar, which means from an academic point of view, there is a lot of inadequate content. More recently, many AI-generated publications and citations' manipulations have also been found to have made their way into Google Scholar. Furthermore, Google Scholar is not well organised for keyword searches. Finally, we considered the 'Newer Dimensions' database. Whilst this has a greater number of documents than Scopus, they are not freely accessible, as they are behind a paywall (see Singh et al. 2021).

Environmental issues to be considered in SEA are listed in Articles 2 to 7 of the UNECE SEA Protocol and Annex I (f) of the SEA Directive. They include '*biodiversity, human health, fauna, flora, soil, water, air, material assets, cultural heritage, including architectural and archaeological heritage, landscape and the interaction among these factors*'. The SEA Directive includes *climatic factors*, and the Protocol refers to *climate*. Furthermore, the SEA Directive includes *population* and the Protocol *natural sites*. The guideline of the EU SEA Directive (European Commission 2004) states that other relevant issues may be included, such as noise.

The aim of the literature search and review was to identify the potential human health impacts associated with environmental issues commonly included in SEA practice. Following the list introduced above, environmental issues to be included in the literature review, therefore, include:

- Air
- Water
- Noise
- Biodiversity/flora and fauna
- Climate
- Soils
- Natural sites
- Landscape/cultural heritage/material assets

As there is a large volume of relevant documents on the environment and human health, restrictions were set to focus the search of literature on the latest and most relevant publications. The search was limited to the documents published within the three years from January 2021 to January 2024, where over 20% of the nearly 80,000 documents from the scopus search mentioned above were published. In cases where there was insufficient information from the articles published in the last three years, the search was extended to five years. For landscape / cultural heritage/ material assets, this had to be extended to 10 years. Furthermore, evidence presented by Pyper et al. (2022) is used when explaining and discussing connections.

The literature search considered the subjects of environmental as well as earth and planetary sciences. It was restricted to review articles, resulting in 10,773 potentially relevant documents being identified.

The search string for identifying documents was designed with reference to the environmental issues listed in Articles 2–7 of the UNECE SEA Protocol and Annex I (f) of the SEA Directive, as explained above. Repeated searches were conducted using the keywords of each environmental subject listed, with 'human', 'health' and 'review'. The following search string was used:

(KEY (*Environmental Subjects* *) AND KEY (human) AND KEY (health) AND KEY (review)) AND (LIMIT-TO (SUBJAREA, 'ENVI' =) OR LIMIT-TO (SUBJAREA, 'EART'))

Documents returned from the search were manually screened in order to select those that were relevant in terms of the subjects and scope of the study.

Using the search strings, we found a total of 2136 papers on Scopus. These were manually screened by reading through the title and abstract. We filtered out those papers that focused on subjects that were outside the scope of SEA. For example, studies on indoor environment, occupational health, geographic comparison, and research methods were excluded. After the screening, a total of 238 papers were further reviewed. Amongst those, many did not provide clear statements about environmental and health interlinkages, and some did not present any empirical support for the statements made. Subsequently, we summarise the findings of the 91 papers that provided clear statements about potential environmental and health linkages. References are provided to all 91 papers in the text. Table 1 in the section 'Synthesis' numbers all references used.

Results

Results of the literature review are summarised with regard to the environmental issues air, water, noise, biodiversity/flora and fauna, climate, natural sites, soils and landscapes/cultural heritage/material assets. The focus of the review is on potential associated health impacts. Implications of the findings for SEA are also briefly explored.

Air

Air pollution is a significant environmental health risk globally. Using the search string described in the methodology above returned 541 documents that were published since January 2021. After screening and removing documents that did not link environmental issues and health outcomes, 79 documents remained for review. Fifteen of those were found to provide for some evidence on interlinkages.

Fine particulate matter ($PM_{2.5}$, PM_{10}), nitrogen dioxide (NO_2) and sulphur (S) are found to be linked to allergies, respiratory diseases, cardiovascular conditions, cancers, kidney malfunctioning, unfavourable pregnancy/birth-related outcomes and premature mortality (Verscheure et al. 2023; Pritchett et al. 2022; Markozannes et al. 2022; Rasking et al. 2022; Song et al. 2023; Sun et al. 2023; Zhu et al. 2024). Furthermore, ammonia (NH_3), as a substantial contributor of fine particulate matter, is associated with respiratory diseases (Wyer et al. 2022). Importantly, $PM_{2.5}$, PM_{10}, ozone (O_3), NO_2, and carbon monoxide (CO) were observed to be positively associated with COVID-19 incidences, in which $PM_{2.5}$ and NO_2 were correlated with COVID-19 deaths (Chen et al. 2024; Hernandez Carballo et al. 2022). Also, air pollution has been connected with obesity in both children and adults (Zheng et al. 2024).

Air pollution has been observed to be associated with mental health problems. Long term exposure to PM, especially PM2.5, for example, has been correlated with cognitive disorders, ADHD, depression and suicide (Thompson et al. 2023; Liu et al. 2021; Davoudi et al. 2021; Zhang et al. 2022). This is important for consideration in SEA even if it is unlikely that quantification is possible.

Key sources of emissions of particulate matter include traffic and industrial activities, as well as mining. SEA routinely focuses on health impacts of air quality in e.g. urban development plans (including transport) and energy infrastructure plans. A challenge for an effective and consistent inclusion of air quality in SEA is a discrepancy between different air quality standards as the basis for predictions on significance. WHO guidelines (WHO, 2021), for example, have thresholds that are more stringent than EU standards. However, they are not legally binding.

Spatial and land use, as well as sectoral PPPs, e.g. those on industrial development and transport, can either exacerbate or mitigate health risks and their associated SEAs can play key roles in reducing potential negative impacts and enhancing positive outcomes. Air quality is routinely addressed in SEA. However, impacts on the health of affected populations, in terms of e.g. number of people and vulnerable groups at risk of particular illnesses as a consequence of air pollution (both, physical and mental) are only rarely attempted (see Pyper et al. 2022).

Water

Access to clean water and adequate sanitation is fundamental to public health. Pollution from industrial discharge, agricultural runoff, and urban waste poses significant risks, particularly to marginalised communities. Scopus found 376 documents concerning water quality and human health published between January 2021 and January 2024. After screening and filtering, 34 documents remained for review. Twelve of those were found to provide for some robust evidence on interlinkages.

Drinking water contaminated by chemicals and microbial pathogens is known to pose various health risks. Chemicals such as residue of fertilisers, herbicides and pesticides are identified to increase the risk of developing cancers (El-Nahhal and El-Nahhal 2021; Syafrudin et al., 2021; Picetti et al. 2022; Derbalah and Sakugawa, 2023). Heavy metals in drinking water can damage different internal organs, increasing the risk of developing disease and cancers (Alagan et al. 2023; Zhang et al., 2023). Furthermore, microbial pathogens such as bacteria and parasites can infect a human body's systems and organs, causing inflammation and chronic diseases (Kristanti et al. 2022; Farrell et al. 2021). Kristanti et al. (2022) also note that there are also concerns that drinking water may play a role in the transmission and outbreak of waterborne diseases.

Antibiotic pollution has been detected in rivers, and antimicrobial-resistant waterborne organisms have been found in coastal and freshwater bodies. This is important, as these have the potential to transmit antimicrobial resistance to humans (Velazquez-Meza et al. 2022; Grenni 2022).

A further concern is plastic pollution in fresh and coastal waters. Microplastics are found to potentially induce gastrointestinal tract inflammation and oxidative stress (Janani et al. 2024). Macro and nano plastics can pass to humans through the food web (Ardestani 2022).

Various aspects of water quality are typically considered in SEA, particularly for those that are prepared for e.g. water management, mining, waste management and transport-related PPPs. In this context, water quality indicators play an important role, including physical (e.g.

temperature and turbidity), chemical (e.g. pH value and pesticide concentration), biological (e.g. bacteria and protozoa) and radiological (e.g. uranium) indicators. In SEA, potential health implications, such as the spread of waterborne diseases, are not always included. An example for a framework that supports the incorporation of water-related health outcomes in SEA includes the WHO's Water, Sanitation, and Hygiene (WASH) initiatives (WHO 2025). These include supporting multi-sectoral technical cooperation and capacity building, provision of data, and integration into disease programmes (e.g. cholera). In the context of SEA, the EU Water Framework Directive (European Parliament 2000) is an important reference for taking water issues into account.

Noise

Sixty-seven documents published between January 2021 and January 2024 were identified in the Scopus search dealing with noise and human health impacts. After filtering them, nine papers were reviewed. All of them were found to provide for some robust evidence on interlinkages. Among these, traffic noise is the most common type of noise, covered in eight of them.

There is evidence for noise exposure being associated with negative physical as well as mental health and well-being impacts, in particular by causing annoyance (Ata Teneler and Hassoy 2023; Seidler et al. 2023). Elevated noise levels at night time have been associated with sleep disturbance (Smith et al. 2022; Ata Teneler and Hassoy 2023). Exposure to noise was found to be associated with some forms of cognitive impairment, which may impact on reading and language skills (Thompson et al. 2022), potentially adversely affecting children's learning (Dohmen et al. 2022; Thompson et al. 2022). There are potential associations between noise and other cognitive functions, such as executive functions (which manage behaviours and self-control), but evidence for this to date has remained thin (Thompson et al. 2022). Noise exposure (mainly from traffic) has been found to affect the central nervous system and brain, contributing to an increased risk of neuropsychiatric disorders (Hahad et al. 2022). Road traffic and railway noise have also been found to increase the risks of diabetes mortality (Vienneau et al. 2024). In this context, noise is likely to be potentially associated with obesity (Gui et al. 2022).

Importantly, noise is an example for an environmental issue where effects may not necessarily be negative. For example, there are health benefits associated with e.g. natural sounds. In this context, Zhu et al. (2024) found that exposure to natural sounds (e.g. water or birds) could benefit health recovery and may enhance cognitive performance.

Key sources of noise with potentially negative health impacts include traffic and industrial activities, as well as other sources, particularly connected with urban environments. SEAs have found to underrepresent noise pollution's health impacts, due to focusing primarily on decibel levels (Pyper et al, 2022). In this context, it is important that perceptions of noise and its disturbance differ in ways that are not captured by decibel measurements. This is connected with e.g. the source of the noise (for example, car traffic is usually perceived to be more disturbing than the noise caused by an approaching and passing train; see UK Health Security Agency, 2023). Furthermore, SEA does not tend to distinguish between affected groups, e.g. vulnerable populations. There are guidelines available that can help addressing shortcomings of current practice. For example, the WHO's Environmental Noise Guidelines (2018) provide actionable metrics. Non-acoustic factors of importance when considering impacts include e.g. gender, age, education, existing general stress levels, duration of stay at dwelling subjected to noise in the day, window orientation of a bedroom or living room towards the street. In the EU, the Noise Directive is an important reference point for SEA.

Biodiversity/flora and fauna

For biodiversity/flora and fauna, in the Scopus search, 181 documents were identified that were published between January 2021 and January 2024. After screening and filtering, 24 were reviewed. Six of those were found to provide for some robust evidence on interlinkages.

Biota and ecosystems have been found to affect human health in multiple ways. Natural and urban ecosystems provide services to human societies, including food production (Marselle et al. 2021a; Giglio et al. 2024), as well as cultural, social and scientific values (Marselle et al. 2021a; Giglio et al. 2024). Urban ecosystems, including green and blue infrastructures have been found to affect mental health, wellbeing, and improved social cohesion (Marselle et al. 2021b; Reyes-Riveros et al. 2021, Robinson et al. 2024; Potter et al. 2023). Here, besides the diversity of species, the abundance of specific distinctive species was found to be associated with improved mental health and well-being (Marselle et al. 2021b).

Biodiversity loss is associated with land conversion and extraction of resources. These activities can have substantial implications for human health and well-being. In connection with the 'one health' concept introduced earlier, changes in biodiversity can be associated with an increased risk of zoonotic diseases. Furthermore, in association with impacts on biodiversity, ecosystem services can be affected, such as clean air and water quality. Whilst SEA is in a very good position to identify and assess linkages, in practice lack of knowledge of biodiversity and health interactions of

those involved in SEA may mean that associated health issues are often not adequately addressed (Pyper et al. 2022).

SEA currently rarely quantifies health impacts of biodiversity loss (Pyper et al. 2022). However, and particularly connected with experiences with the COVID-19 pandemic, there has been an increasing awareness of the 'one health' approach, which integrates human, animal, and ecosystem health. This can provide a holistic framework for SEA (Fischer and Cave 2018).

Climate

Regarding an association between climate and human health, 341 documents were identified by the Scopus search between for the period of January 2021 to January 2024. After screening and filtering for relevance on direct linkages being made between climate change and health, 33 documents were reviewed. Nineteen of those were found to provide for some robust evidence on interlinkages.

An immediate direct impact of climate change is a change in temperatures. Heat is related to ill health, such as dehydration (Faurie et al. 2022). Furthermore, an increase in temperatures is found to elevate the risk of cardiovascular diseases (Faurie et al. 2022; Jurgilevich et al. 2023), kidney diseases (Liu et al. 2021a; Faurie et al. 2022, Jurgilevich et al. 2023), respiratory diseases (Faurie et al. 2022; Z. Zhu et al. 2025), type 2 diabetes and mental disorders (Natur et al. 2022). Heat also increases the morbidity risks of diabetes patients (Moon 2021). Heat has the potential to have adverse effects on pregnant women, increasing the risk of unfavourable birth outcomes (Haghighi et al. 2021; Dalugoda et al. 2022; Syed et al. 2022). Elevated temperatures are found to increase the toxicity of nano-plastics to microorganisms and aquatic animals, but effects on humans remain uncertain (Zhang et al. 2022a).

In addition to heat, extreme weather events and changes in the natural environment are associated with climate change. These also pose multiple health risks to humans. Heatwaves increase the risks of cardiovascular disease, kidney diseases (Jurgilevich et al. 2023), and mental disorders (Li et al. 2023a). Storms and flooding cause accidents and injuries, which can also affect mental health, exacerbating any existing physical and mental health conditions (Anderson et al. 2022; Jurgilevich et al. 2023; Patwary et al. 2024). Wildfires emit particulate matter, which is found to increase the negative effects of non-wildfire particulate matter (Jiao et al. 2024). Respiratory and cardiovascular diseases increase healthcare needs in the aftermath of extreme weather events that lead to, for example, floods, droughts and wildfires (Lee et al. 2023).

Studies found that ocean acidification and warming increase the toxicity of algae blooms and associated harmful effects on humans (Dermawan et al. 2022). Climate change is suggested to affect individual susceptibility to Tuberculosis, particularly among developing countries (Kharwadkar et al. 2022). Among tropical low and middle-income countries, it is also suggested that the transmission rates of several vector-borne diseases may increase (Filho et al. 2022). Children have been found to be more vulnerable to climate-related disease burden (Weeda et al. 2024).

SEA can play a key role for consideration of climate in e.g. national, regional and local spatial and land use PPPs. In this context, it can assess aspects such as the urban heat island effect and suggest mitigation measures. In other sectoral PPPs it can establish potential carbon emissions of different development options and identify how e.g. disadvantaged or vulnerable groups contribute to emissions and whether they are potentially affected to a significantly greater extent than the rest of the population (Fischer et al. 2011; Pyper et al. 2022).

Soils

The Scopus search resulted in 551 papers covering potential health risks between soils and human health. After filtering for determining direct links between soils and health being made, 14 were reviewed. All of them were found to provide for some robust evidence on interlinkages.

Publications covered the contamination of soils through agricultural activities, mining and the accumulation of microplastics in soil. While contaminated soils from industrial incidences are known to pose significant health risks to humans (as established in e.g. Seveso sites documents; European Parliament 2012), none of the reviewed articles focused on this. It has been suggested that there is a direct connection between soil health and human health, as soil provides multifunctional services to humans, such as food production, nutrient cycling, antibiotics and soil-borne pathogens (Kopittke et al. 2024).

Contaminated soil can affect health directly through ingestion, inhalation and dermal contact. It can also affect human health indirectly, e.g. through consumption of contaminated agricultural products, or water contamination caused by surface runoff (Liu et al., 2023; Nuruzzaman et al., 2025).

Exposure to chemicals used in agricultural activities (e.g. fungicides and pesticides) can cause various illnesses. Pesticides are associated with acute and chronic toxicity, with a potential to cause neurodevelopmental birth defects, asthma, and cancers. There are potential associated long-term health impacts, particularly on children (Kaur et al. 2024). Fungicides (e.g. triazoles) can be toxic and can have adverse effects on the liver, heart, endocrinal, developmental and reproductive systems (Bian et al. 2025).

Mining activities have been associated with heavy metals and potentially toxic trace elements, which are found to pose carcinogenic and non-carcinogenic risks to humans. They affect the kidneys, the liver and skin and include e.g. neuralgia and gastroenteritis. Heavy metals pose higher risks to children than to adults (Hou et al. 2023; Setu and Strezov 2025).

Pharmaceutical contamination is highlighted to be an emerging issue. Pharmaceuticals can enter soil through livestock manure, wastewater, animal excrement and waste disposal (Gentile et al. 2024; Sardar et al. 2025). Pharmaceuticals can also be taken by plants and enter into the food chain. They have been observed to accumulate and can subsequently lead to antibiotic resistance, endocrine disruption, neurological and mutagenic effects (Sardar et al. 2025).

Several papers considered potential health risks associated with microplastics in soils. These are associated with litter, agricultural plastics, sewage sludge, improper sterilisation, plastic mulch films, e-commerce, airborne plastic decomposition, and marine debris (Nath et al. 2024). Microplastics can traverse biological barriers (e.g. intestinal wall and blood-brain barriers) and thus affect vital organs and tissues (Chang et al. 2024). Bioaccumulation of microplastics in the human body can cause inflammation, allergic reaction, oxidative stress, and can induce e.g. respiratory, digestive, urinary, endocrine, nervous, circulation and reproduction issues (Garai et al. 2024; Lee et al. 2024; Shi et al. 2024).

SEA can play a key role in identifying e.g. contaminated areas and sites in spatial and land use PPPs (Fischer 2021). If conducted for new industrial sites, by taking into account potentially produced products, it can identify, early on, mitigation strategies and options for avoiding potential contamination. There are limits to which SEA can take the issues of pharmaceuticals and microplastics into account, though, as these are connected with a range of policy and other decisions that are not associated with spatial and sectoral PPPs that SEA would typically assess.

Natural sites

The UNECE's implementation guide of the Aarhus Convention states that "Natural sites may refer to any objects of nature that are of specific value" (Ebbesson et al. 2014, p.51). The health benefits associated with the provision of accessible natural sites have caught much research attention in recent years. The Scopus search identified 52 documents, published between January 2021 and January 2024. After screening and filtering them, 39 were reviewed. Ten of those were found to provide for some robust evidence on interlinkages. Most focused on green spaces, with a few also mentioning blue spaces.

Green and blue spaces provide human health benefits in multiple ways. Generally speaking, they benefit human health with regard to five groups of functions (Zhao et al. 2022; Potter et al. 2023):

- Ecosystem services;
- Absorption of pollutants;
- Spaces for physical activities;
- Alleviation of stress; and
- Enhancement of sleep quality.

Green spaces are found to improve human mental health, can encourage physical activities and result in cardiovascular benefits (Lévesque-Vézina and Lapointe 2024). Green spaces enable an exposure of calming natural environments that can reduce anxiety (Labib et al. 2022) and lower the risk of depression (Liu et al. 2021a; Labib et al. 2022, Bolanis et al. 2024), as well as reducing noise-induced stress (Ferrante et al. 2025). Provision of green spaces has also been found to enhance social cohesion (Huang and Lin 2023) by offering spaces to engage in daily life activities (Huang and Lin 2023). Letting children spend time in green spaces is suggested to help children's intra- and interpersonal socio-emotional functions and development (Mygind et al. 2021). Exposure to green space has been suggested to have protective effects on non-communicable disease hospital admissions (Luque-García et al. 2022).

SEAs play an important role in helping to avoid or mitigate negative impacts on natural sites and green spaces and enhancing and expanding existing ones (Fischer et al. 2018). This applies to various types of PPPs, including transport, spatial/land use, agricultural, mining, waste, water management and other PPPs. Furthermore, SEA can be applied to development PPPs of natural sites and green spaces.

Landscape/cultural heritage/material assets

The Scopus search identified 15 documents that dealt with connections of landscape and human health, published between January 2021 and January 2024. No documents were identified with regards to connections between cultural heritage or material assets and health. Due to the low number of documents, the search was extended to include documents published within ten years (i.e. from 2013). With the extended search, 27 documents were identified. Eight of those were found to provide for some robust evidence on interlinkages.

Six documents focused on natural landscapes. There is therefore overlap with natural sites. These state that natural landscapes provide ecosystem services to humans (Coutts and Hahn 2015; Nishi and Hashimoto 2022). Early-childhood exposure to diverse microbiota has also been

said to have positive effects on a healthy development (Matthews et al. 2024). Similar to green spaces, exposure to natural landscapes is found to benefit human well-being and to improve mental health, as well as providing spaces for physical activities (Zhang et al. 2021; Hjort et al. 2023; Huang and Lin 2023; Li et al., 2023b). Natural environments have been found to help patients on psychiatric wards to recover (Hjort et al. 2023). In an urban landscape, transportation and the built environment characteristics (e.g. connectivity, aesthetic, safety) affect adults' motivations for physical activities (Salvo et al. 2018).

The important role SEA can play with regards to landscapes is similar to that identified for natural sites. At times SEA has also been applied to landscape PPPs (Hanusch and Fischer 2011) and development of parks in urban areas can be accompanied by SEA, too.

Synthesis—Table on environmental-related health risks

Table 1 synthesises findings with regard to the linkages between environmental changes and potential health impacts. These are expressed here as health risks. In addition, vulnerable groups are also listed. Here, 'vulnerable' means more likely to have health impacts/more severe health impacts than the general adult population. Commonly referred to groups in this context include pregnant women and children. The key reason is that the immune system of foetuses, babies and children are underdeveloped, and that the health of mothers is important for the development of them. Another frequently mentioned group is the elderly. The likelihood of exposure is not included in the table. In this context, issues such as poverty or cultural background may be of relevance. Other vulnerable groups include those with pre-existing conditions, including e.g. those with asthma, diabetes or cancer. All environmental changes can potentially affect those with pre-existing conditions. They are therefore not specifically mentioned in the table.

Challenges and opportunities for an effective integration of health in SEA

Despite knowledge of the numerous connections between environmental impacts and health, practical integration of health into SEA has faced important challenges. These include the availability of only limited guidance documents, data issues (gaps, data protection; see e.g. Dallhammer et al. 2019), the existence of disciplinary silos and resource constraints (Pyper et al. 2021). Currently, most SEA guidelines offer very limited detail on how to assess health impacts (Pyper et al. 2022). This usually leaves those conducting SEA uncertain about what approach to adopt, and what indicators and data sources to use. Reliable data for specific health indicators are frequently not readily available or can only be accessed in fragmented ways (Dallhammer et al. 2019). Environmental and health experts and practitioners are often based in different administrations/departments and do not usually work together. This limits opportunities for collaboration. As a result, health sections in SEA may not integrate well with sections on other environmental issues (Fischer et al. 2010). Finally, on many occasions, SEAs are subject to tight budgets and timelines (Fischer et al. 2023). These are all barriers for a comprehensive consideration of health aspects in SEA. We hope that our paper can be the starting point for addressing challenges.

Opportunities for an effective consideration of health in SEA can be associated with approaches introduced by the WHO and the UNECE. The WHO supports those engaged in SEA with various guidelines. These include e.g. 'Environmental Noise Guidelines for the European Region' (WHO 2018) and 'Global Air Quality Guidelines' (WHO 2021). These documents provide evidence-based thresholds for environmental determinants that influence health outcomes. They are useful resources for an integration of health into SEA.

The UNECE (2022) introduced the DPSEEA (Drivers, Pressures, State, Exposure, Effects, Actions) Framework, which can serve as a tool for tracing environmental changes to health outcomes. In this context, drivers are understood to be socio-economic factors (urbanisation, economic development) that initiate environmental changes. Furthermore, pressures are changes to environmental conditions, which are triggered by drivers. Examples of pressures include air pollution and habitat loss. The state focuses on the altered state of the environment as a consequence of the pressures applied. Typical examples include degraded air and water quality. Explicit linkages with health are made through the consideration of exposure, which describes the interaction of populations with altered environmental conditions. Effects are health outcomes arising from the exposure. Examples for outcomes include respiratory illnesses and cardiovascular conditions. Finally, actions are the interventions that aim at mitigating risks and at enhancing benefits. Examples include changes to PPPs and observed improvements in environmental and health conditions and outcomes.

Strengths and weaknesses of the methodology underlying the paper

Whilst the methodological approach underlying this paper has various strengths, there are also some potential

Table 1 Environmental related Health risks and vulnerable groups

Environmental aspect	Health risks	Vulnerable groups [not exhaustive]
Air	• Cardiovascular and respiratory disease risks (Markozannes et al. 2022 **[1]**; Sun et al. 2023 **[2]**; Wyer et al. 2022 **[3]**); cancers (Pritchett et al. 2022 **[4]**); other diseases and body functions (Rasking et al. 2022 **[5]**); viruses (Chen et al. 2024**[6]**) • Allergies (Verscheure et al. 2023 **[7]**) • Unfavourable birth outcomes (Markozannes et al., 2022, Song et al. 2023**[8]**, Zhu et al. 2023) **[9]** Mental Health issues (e.g. depression and anxiety, ADHD) (Zhang et al. 2022**[10],**; Davoudi et al. 2021 **[11]**, Liu et al. 2021b **[12]**) • Cognitive disorders (Thompson et al. 2023**[13]**) • Development of obesity (Zheng et al. 2024 **[14]**) • Deterioration of conditions for those with existing respiratory diseases (Hernandez Carballo et al. 2022 **[15]**, Verscheure et al. 2023)	Children, pregnant women, elderly
Water	• Non-communicable disease risk (damage to internal organ and cancer risk) (Janani et al. 2024 **[1]**; El-Nahhal and El-Nahhal 2021 **[2]**; Syafrudin et al. 2021 **[3]**; Picetti et al. 2022 **[4]**; Ardestani, 2022 **[5]**; Derbalah and Sakugawa 2023 **[6]**; Zhang et al. 2023 **[7]**) • Microbial infections (Kristanti et al. 2022 **[8]**) • Waterborne disease transmissions (Alagan et al. 2023 **[9]**) • Antimicrobial resistance (Farrell et al. 2021**[10]**, Grenni 2022 **[11]**; Velazquez-Meza et al. 2022 **[12]**)	No vulnerable groups specified
Noise	• Sleep disturbance (Smith et al. 2022 **[1]**, Ata Teneler and Hassoy 2023 **[2]**; Dohmen et al. 2023 **[3]**; Seidler et al. 2023 **[4]**; Zhu et al. 2024 **[5]**) • Cognitive functions (Thompson et al. 2022 **[6]**) • Impact on central nervous system and brain (Hahad et al. 2022 **[7]**) • Adiposity (excessive fat issue in the body) (Gui et al. 2022 **[8]**) • Diabetes (Vienneau et al. 2024 **[9]**)	Children
Biodiversity / flora and fauna	• Changes to immune and digestive system (contamination of food and microbial resistance) (Marselle et al. 2021a **[1]**, Marselle et al. 2021b **[2]**; Potter et al. 2023 **[3]**; Robinson et al. 2024**[4]**) • Health issues connected with imbalanced diets (Marselle et al., 2021a, Marselle et al. 2021b, Giglio et al. 22024 **[5]**) • Mental health and well-being (Marselle et al. 2021a, Marselle et al. 2021b; Reyes Riveros et al. 2021 **[6]**; Potter et al. 2023; Giglio et al. 2024; Robinson et al. 2024)	No vulnerable groups specified
Climate	• Non-communicable disease risks associated with heat (e.g. respiratory, cardiovascular and kidney disease; waterborne diseases) (Liu et al. 2021 **[1]**, Faurie et al. 2022**[2]**; Zhang et al. 2022 **[3]**; Jurgilevich et al. 2023 **[4]**; Zhu et al. 2025 **[5]**) • Gastrointestinal problems (Dermawan et al. 2022 **[6]**; • Increases in communicable diseases (e.g. tuberculosis)(Kharwadkar et al. 2022 **[7]**) • Increases in virus, bacteria, parasites' and other related illnesses (Filho et al. 2022 **[8]**) • Unfavourable birth outcomes (Haghighi et al. 2021 **[9]**; Dalugoda et al. 2022 **[10]**; Syed et al. 2022 **[11]**) • Injuries, mental health risks and an exacerbation of existing health conditions due to extreme climate events (flooding, heatwaves and wildfires)(Anderson et al. 2022 **[12]**, Natur et al. 2022 **[13]**; Jurgilevich et al. 2023; Li et al. 2023 **[14]**; Jiao et al. 2024 **[15]**; Patwary et al. 2024 **[16]**; Weeda ert al. 2024 **[17]**; Moon, 2021 **[18]**) • Impacts on health services (Lee et al. 2023 **[19]**)	Pregnant women, children, the elderly, diabetes patients
Soils	• Carcinogenic and non-carcinogenic risks (e.g. liver, heart, skin endocrine, neuralgia, gastroenteritis and reproductive systems; connected with nitrates, toxins, microplastics and others) (Bian et al. 2025 **[1]**; Hou et al. 2023 **[2]**; Liu et al. 2023 **[3]**; Setu and Strezov 2025 **[4]**; Chang et al. 2024 **[5]**; Garai et al. 2024 **[6]**; Lee et al. 2024 **[7]**; Nath et al. 2024 **[8]**; Kaur et al. 2024 **[9]**; Shi et al. 2024 **[10]**; Nuruzzaman et al. 2025 **[11]**)	Children

Table 1 (continued)

Environmental aspect	Health risks	Vulnerable groups [not exhaustive]
	• Antibiotic resistance endocrine disruption, neurological effects and mutagenic effects (Sardar et al. 2025 **[12]**; Gentile et al. 2024 **[13]**; Kopittke et al. 2024 **[14]**)	
Natural sites	Lack of green spaces deprives populations of (Mygind et al. 2021 **[1]**; Labib et al. 2022 **[2]**, Zhao et al. 2022**[3]**) Luque-Garcia et al. 2022 **[4]**; Bolanis et al. 2024 **[5]**; Liu et al. 2023 **[6]**; Potter et al. 2023 **[7; also mentioned under 'Flora/Fauna']**; Lévesque-Vézina and Lapointe 2024 **[8]**; Huang and Lin 2023 **[9]**; Ferrante et al. 2025 **[10]**): • mental health benefits • opportunities for physical activities • respiratory and cardiovascular benefits • opportunities to reduce anxiety, stress, risk of depression	Children, the elderly
Landscape	***See natural sites as well as biodiversity/flora and fauna above*** Also, natural landscapes provide ecosystem services to humans (Coutts and Hahn 2015 **[1]**; Salvo et al. 2018 **[2]**; Zhang et al 2021 **[3]**; Nishi and Hashimoto 2022 **[4]**; Hjort et al. 2023 **[5]**; Huang and Lin 2023**[6; also mentioned under 'Natural Sites']**; Li et al. 2023 **[7]**; Matthews et al. 2024 **[8]**)	Children, the elderly

NB Bold numbers in square brackets show the references identified in the literature review

weaknesses. Both are briefly outlined here. Our review captures the most recent developments in the field of environmental–health interactions and therefore provides a timely summary of current knowledge. Also, by focusing on those environmental aspects covered by the SEA Directive and the SEA Protocol, actionable insights are created for those engaging with SEA in practice. This addresses the gap between academic research and practical application. Importantly, evidence-based recommendations are provided.

On the other hand, focusing on published papers listed in Scopus means that possible insights from the grey literature (e.g. government reports) have not been considered. Also, only English-language publications were included. Any region-specific findings in non-English formats were therefore shunned. Furthermore, methodologically, we filtered the Scopus search with keywords and subject areas. As different journals set their keywords differently, and Scopus tends to assign subject areas based on the journal rather than article content, papers from certain journals may not have appeared, even if they are about environmental health. Finally, the identification of vulnerable groups was generalised from an environmental science rather than a medical science point of view.

Conclusions

SEA frameworks, such as the European SEA Directive and the UNECE SEA Protocol, explicitly include health as a key aspect for assessment. However, practical implementation is frequently lagging behind intentions. This is even true when health is considered only through the lens of direct bio-physical environmental impacts. In order to address this issue, in this paper, the authors have provided for a systematic and comprehensive literature review on how various bio-physical environmental aspects that are assessed in SEA are interrelated with numerous potential human health impacts.

Bio-physical environmental aspects to be considered according to the SEA Directive and the UNECE SEA Protocol include air, water, flora and fauna (biodiversity), soils, the climate, natural sites, landscapes cultural heritage and material assets. A wide range of potential physical as well as mental health effects are interrelated with these bio-physical environmental aspects. According to the professional scientific literature, these include risks of (microbial) infections, communicable (e.g. bacteria, viruses, fungi or parasites) and non-communicable (e.g. cardiovascular and respiratory, cancers and diabetes) diseases, risks to cognitive functions, unfavourable birth outcomes, and development of obesity. Children,

pregnant women, the elderly and those with pre-existing health conditions may be more seriously affected by certain environmental impacts.

The proportionate consideration of health in SEA is important. Proportionality focuses on a comprehensive assessment of potential health impacts, while acknowledging constraints set by the context within which PPP making is happening. Significant health impacts that are of relevance for a particular PPP need to be prioritised. In this context, issues such as the administrative level, the systematic tier, institutional capacity and power relationships that are present in a particular PPP situation need to be considered in order to be able to conduct assessments are both, rigorous and feasible.

A key challenge for a proportionate consideration of health in SEA has been found to be the absence of specific guidelines. The findings presented in this paper should therefore be useful to those engaging with health in SEA.

Data availability

No datasets were generated or analysed during the current study.

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Author contributions Thomas B Fischer wrote the main manuscript; Ben Cave was a project partner and contributed to the literature review and the drafting of the manuscript; Hung Shiu Fung conducted large parts of the literature Review; Tiago Rodrigues contributed parts to the writing of the manuscript; Birgitte Fischer-Bonde was involved in the literature Review; Ainhoa Gonzalez was the Project Lead and provided detailed editorial input to the manuscript

Compliance with Ethical Standards

Competing interests The authors declare no competing interests.

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